



MORK MONGKUL

Data Scientist | AI Engineer | Data Engineer



096 309 86 72



morkmongkul369@gmail.com



<https://www.morkmongkul.dev>

EDUCATION

Bachelor of Data Science in Engineering

11/03/2022-2/02/2026

Graphic Design and Media

Instinct Institute

2022-2023(Graduate)

BAC II(National Exam)

Hunsen Kransamor High School

2018-2021

CERTIFICATION & EXTRA

- Data Scientist Nanodegree Program (**Udacity**)
- Google Data Analytics (**Coursera**)
- SQL and Relational Databases 101 (**Conigitive**)
- Microsoft Power BI Data Analyst (**Coursera**)
- Data Warehouse Concepts, Design, and Data Integration(**Coursera**)
- Machine Learning Specialization 2022(**Coursera**)
- UNESCO UNITWIN Data Science Camp (**UNECO UNITWIN and Handong Global University**)
- Career-Ready Web Developer(**FrontendMaster**)
- The Frontend Developer Career Path (**Scrimba**)

TECHNICAL SKILLS

- **Programming:** Python, JAVA, SQL, R, HTML/CSS, Javascript, Typescript, nodejs.
- **Libraries:** Pandas, NumPy, OpenCV, Pillow Matplotlib, Seaborn, Scikit-learn, Pytorch...
- **Tools:** Power BI, Tableau, Excel, Jupyter Notebook
- **Database:** MySQL, SQLite, SQL Server
- **Other:** Git, Canva, Figma, Adobe PS, AI, PR, AE, Cinema 4D

LANGUAGE

- Khmer -**Native**
- English -**B2**
- French -**A2**

PROFILE

I am a Data Science and Engineering graduated with a creative background in graphic design and digital marketing. My transition into the data field is backed by proficiency in Python, SQL, data visualization (Power BI, Tableau), and a passion for uncovering insights through data. With hands-on experience in analytics, reporting, and campaign optimization, applied Machine Learning and Deep Learning to solve business problems. I am always eager to learn with new opportunity to further develop my skills in Data Science, Data Analytics, Business Intelligence, or Data Engineering.

WORK EXPERIENCE

Data Scientist

Cambodia Development Resource Institute (CDRI) 2025-Present

- Contributed to the ongoing development and enhancement of the institutional data platform.
- Maintained data platform servers and integrated automated data pipelines.
- Designed and implemented scalable, reliable, and secure data infrastructure.
- Developed interactive data visualizations to support efficient data access and retrieval.
- Collected, cleaned, and analyzed data from multiple internal and external sources.

Artificial Intelligence Bootcamp Instructor

Instinct Institute 2025-Present

- Prepare a high quality teaching materials for the entire AI-Boocamp
- Provide students a clear, detailed, and engaging learning experience.
- Teaching covers, Python for DS, Machine Learning, Capstone Project and Deployment.
- Always happy to share and giving mentors to others.

Research Data | Data Engineer & AI engineering-Intern

Cambodia Development Resource Institute (CDRI) 3 months

- Assisting in developing CDRI data-hub website
- Helping in designing a database for CDRI datahub, data catalogue.
- Making Interactive data visualization of CDRI website.
- Integrating AI chatbot assistant for CDRI datahub website.

Machine Learning Engineer

Team Lead of Applied ML in Agriculture Lab(ReDa-Lab) - 2025- present

- Lead applied machine learning initiatives for agriculture-focused research platforms.
- Design and develop scalable data systems, including databases and data catalogs.
- Build interactive data visualizations to improve insight and accessibility.
- Integrate AI chatbot assistants for intelligent data discovery and user support.
- Collaborate with researchers and engineers to deliver ML-driven solutions.

MOST RECENT PROJECTS

Khmer Name Entity Recognition

Trained from scratch at character-level using RNN(GRU) architecture for Khmer language understanding, use this as model to pre-trained for Name Entity Recognition task with Bi-LSTM-CRF for entity extraction task. This project aim to give the model the capability to detect entity such as LOCATION, NAME, ORG, DATE-TIME..., achieved the 98% precision for entity recognition on testing.

Khmer Next Word Prediction

Implemented statistical model(N-gram), to predict the next word probability of the current user typing in real-time, this model perform well with **perplexity < 100**, suggesting that the model perform well in word prediction, this project extend to fine-tuning the LLM(Qwen-8B) for more better khmer word prediction in the future.

Chat With Document-Chatbot

A personal chatbot which give you can upload your data in various format such as pdf, docs, csv or excel. You can input questions about your documents and get a real-time answer with the context of your documents.

Khmer legal Chatbot Assistant

A Legal chatbot specialize for Cambodia Law, including labor law, criminal, civil, administrative, and commercial law and any regulations documents that frequently update. This chatbot capable to answer user queries in both Khmer/English language, provide a clear information about law, citations, and referencing.

CDRI Data Hub

CDRI DataHub is a public knowledge platform that brings together trusted development data, indicators, and visual insights to support evidence-based planning and policy in Cambodia.

Enhancing Khmer Sentiment Analysis using Transformer Model

Fine-tuned multilingual transformer models (**mBERT**, **XLM-RoBERTa**) on a custom-labeled Khmer dataset to improve sentiment classification. Focused on data preprocessing, tokenization, and model evaluation, addressing challenges in low-resource language processing.

Customer Churn Analysis

Performed exploratory data analysis and feature engineering on customer behavior data to predict churn using logistic regression and random forest. Delivered actionable insights to improve customer retention strategies.

Object Detection and Localization

Developed a computer vision system using YOLOV5 for real-time object detection. Trained on custom datasets and evaluated model precision, recall, and bounding box accuracy.

Job Search Data Visualization

Created an interactive dashboard using Tableau,Power BI and Python to visualize job market trends, salary ranges, and keyword frequency. Enabled better career planning through data storytelling.

Data Warehouse Design

Designed a scalable data warehouse schema for a retail business using dimensional modeling. Implemented ETL processes to integrate data from multiple sources for business intelligence reporting.